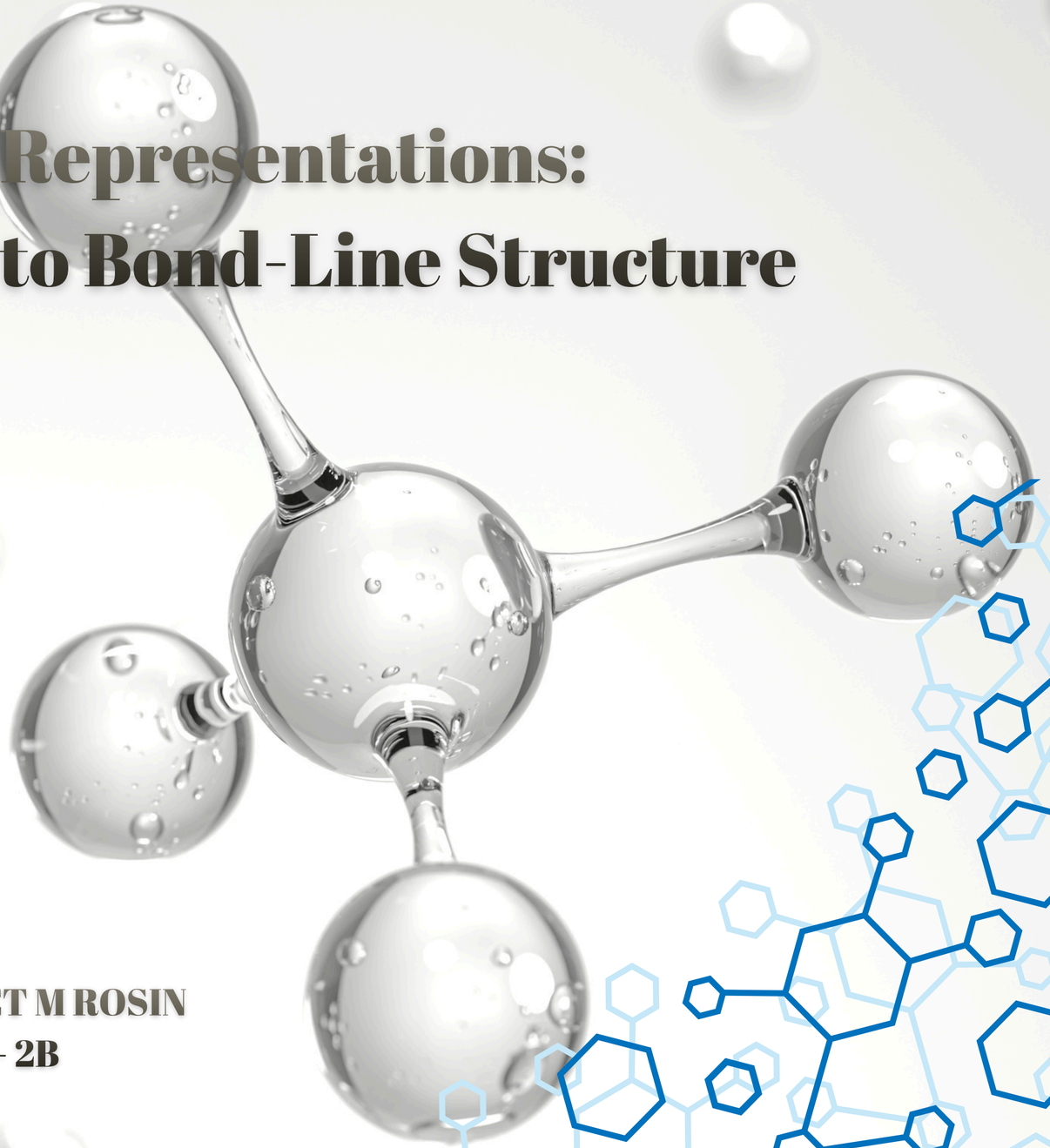
One Molecule, Four Representations:
From Molecular Formula to Bond-Line Structure



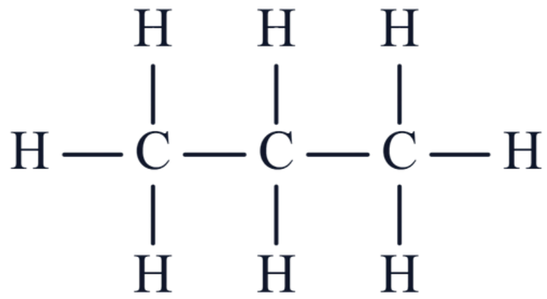
PAUL BENEDICT M ROSIN
BSChE - 2B

PROPANE

MOLECULAR FORMULA

$$C_3H_8$$

LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2CH_3$$

BOND-LINE FORMULA

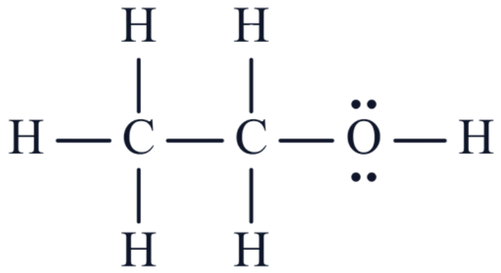


ETHANOL

MOLECULAR FORMULA

$$C_2H_6O$$

LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2OH$$

BOND-LINE FORMULA

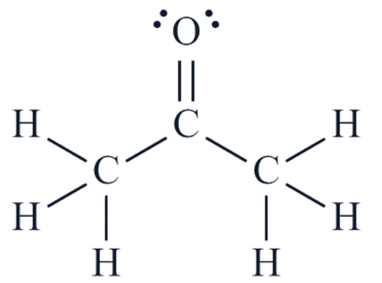


ACETONE (PROPAN-2-ONE)

MOLECULAR FORMULA

$$C_3H_6O$$

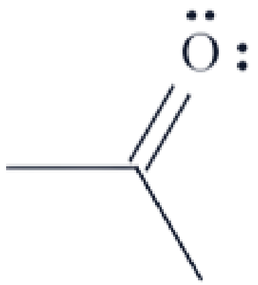
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)CH_3$$

BOND-LINE FORMULA

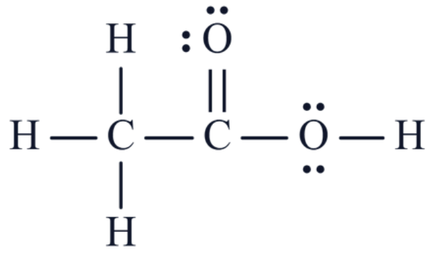


ACETIC ACID (ETHANOIC ACID)

MOLECULAR FORMULA

$$C_2H_4O_2$$

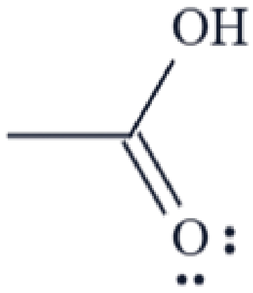
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$CH_3COOH$$

BOND-LINE FORMULA

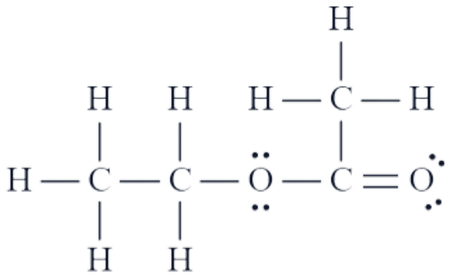


ETHYL ACETATE

MOLECULAR FORMULA

$$C_4H_8O_2$$

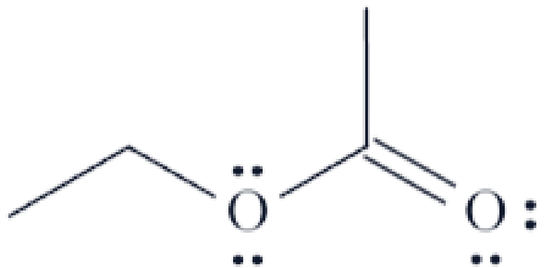
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$CH_3COOH_2CH_3$$

BOND-LINE FORMULA

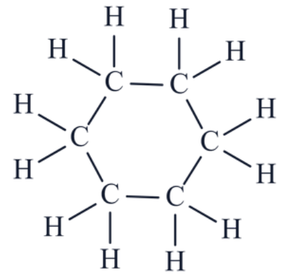


CYCLOHEXANE

MOLECULAR FORMULA

$$C_6H_{10}O$$

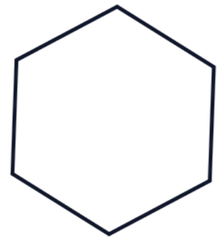
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$CH_2CH_2CH_2CH_2CH_2CH_2$$

BOND-LINE FORMULA

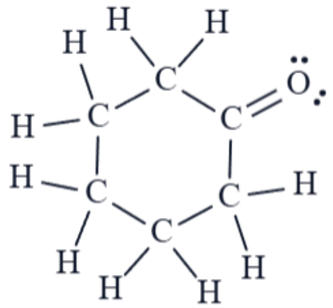


CYCLOHEXANONE

MOLECULAR FORMULA

$$C_6H_{12}$$

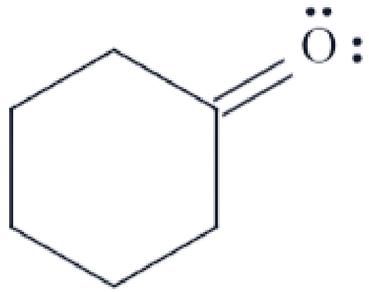
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$(CH_2))_5CO$$

BOND-LINE FORMULA

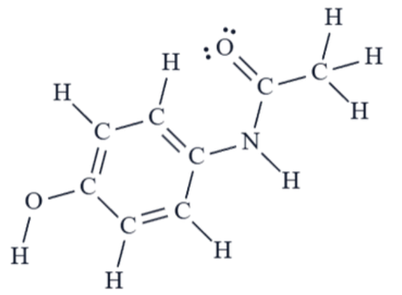


ACETAMINOPHEN (PARACETAMOL)

MOLECULAR FORMULA

$$C_8H_9NO_2$$

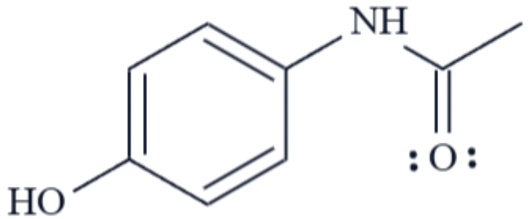
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$HOC_6H_4NHCOCH_3$$

BOND-LINE FORMULA

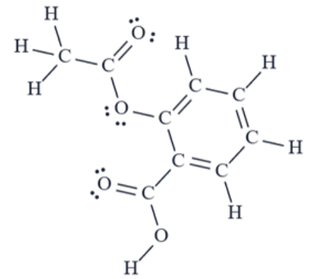


ASPIRIN (ACETYLSALICYLIC ACID)

MOLECULAR FORMULA

$$C_9H_8O_4$$

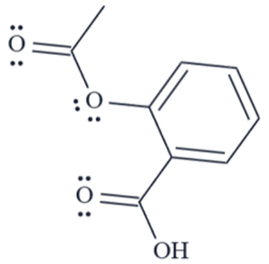
LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$CH_3COOC_6H_4COOH$$

BOND-LINE FORMULA

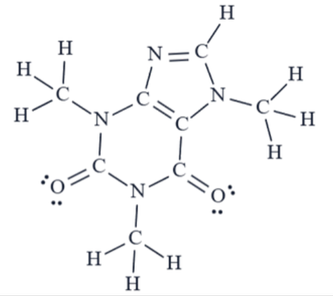


CAFFEINE

MOLECULAR FORMULA

$$C_8H_{10}N_4O_2$$

LEWIS STRUCTURE



CONDENSED STRUCTURAL FORMULA

$$CH_3N_4C_5O_2(CH_3)_2$$

BOND-LINE FORMULA

